

DESIGN PARKING LOT

Overview

Req. Gathering

Design Patterns

Class diagram

Schema diagram

code } - H-CO
APP }

* Watch 1st 2 videos

* Notes and other resources are
in GitHub.

Overview

- Align with interviewer
- Try to understand what interviewer is asking.

Design parking lot

you know the system

you mean, which we can see in malls, airports, etc. where we can park the vehicle, is this what you want me to design?

you don't know the system

Thank you for your question. Sorry I am not familiar with the parking lot. Could you please provide a brief overview of it?

① Do you want me to ^{Req. Gathering}

① design entities // OR

② Complete software system.

② Where can we store the data?

① In-memory OR

② In real database.

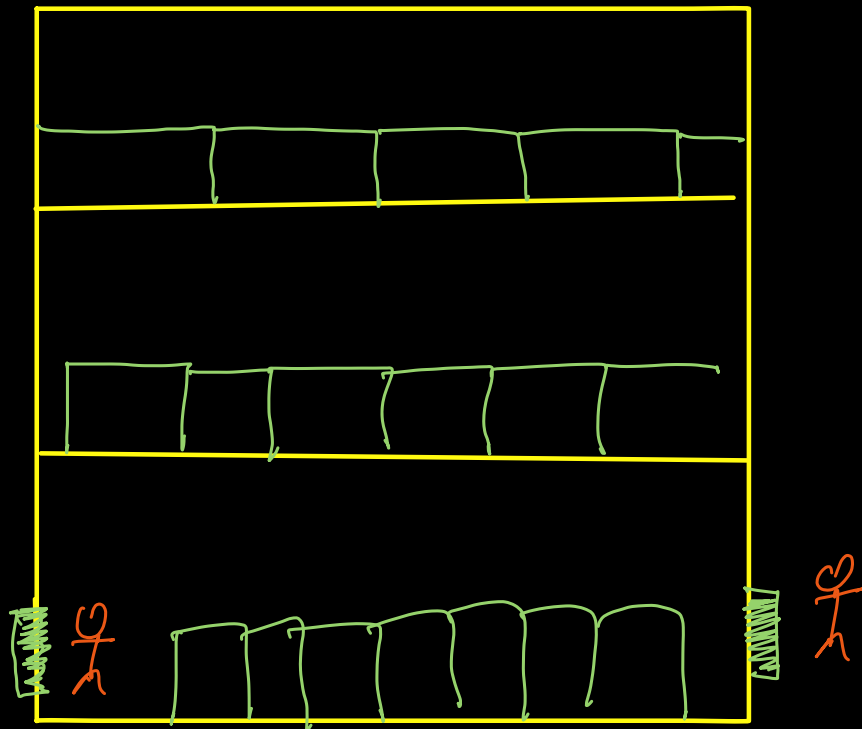
③ How user can interact with the system?

- ① API OR
- ② CMD OR
- ③ Hard code

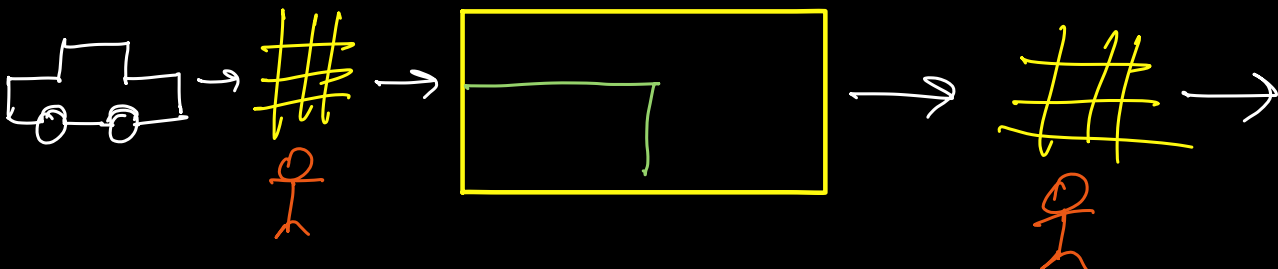
REQUIREMENT GATHERING & DOUBT CLEARING

- ① Visualization
- ② User journey

Visualization



User Journey



1. Parking lot has multiple floors.
2. Different types of vehicle can be parked
3. Multiple entry and exit gates for parking lot.
4. Each floor will have multiple parking spot.
5. it'll be a human operated parking lot management system.
6. At the time of entry, a spot will assign to the vehicle, and a ticket will assign to vehicle owner.
7. Parking fee calculation can be changed based on
 1. Vehicle Type
 2. Time
 3. Vehicle Type + Time
8. System should support both online and offline payments

Design Patterns

- ① Fee Calculation → Strategy Design
 - Vehicle based
 - Time based
 - Vehicle and time based
- ② Payment → Adapter design
 - Online
 - Offline